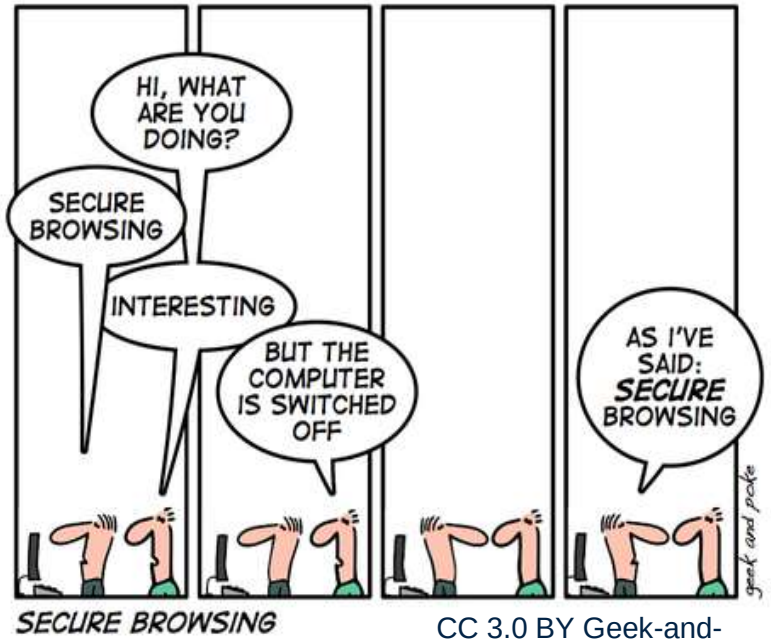
Week 2 Control Frameworks

- Controls are security requirements
  - Ransomware protection: Controlled Folders Access
- Frameworks help us select controls
  - ACSC Essential 8, e.g. App control using WDAC
  - CIS Controls = Essential 8 + more, e.g.
    - CIS 9.2 DNS filtering & CIS 9.5 DMARC
- PowerShell
  - Common info model, e.g. installed SW: `Get-CimInstance Cim_Product`
  - `Get-WinEvent system`
  - `Get-Process`, e.g. rogue processes
  - `Get-Service`, e.g. firewall running?
  - `Invoke-WebRequest`…
  - `ls env:`
  - `ls HKCU:`



CC 3.0 BY Geek-and-Poke

Controls and Frameworks

- Controls are like CIA requirements or goals (NIST 2020, Kohnke, Shoemaker & Sigler 2016)
  - Many ways to implement, e.g. "Implement automated backups"
    - Little implementation guidance, e.g. to avoid promoting any organisation
    - Audience often assumed to be security experts
  - Types of Controls
    - Technical controls, e.g. firewall, ACLs, VPNs, IDS/IPS
    - Admin controls, e.g.
      - Policies, e.g. acceptable use, password & data retention policies
      - Training requirements
      - Response plans required to be created
    - Physical controls, e.g. locks & security cameras
- Control Frameworks are collections of controls
  - Help us select & prioritise our actions
  - Comprehensive but overwhelming, e.g. NIST CSF & NIST SP 800
  - Prioritised subsets to help us start, e.g. Essential 8 & CIS Controls

ACSC Personal Cyber Security Controls

- Australian Cyber Security Centre (ASCS)
  - First step controls

Automated updates of devices & software	Step-by-step guides
Use MFA, e.g. authenticator app, to protect your accounts	Step-by-step guides
Automated backups	Step-by-step guides
Secure mobiles, e.g. autolocks, remote locking & wiping; turn off unused networking	
Passphrases	Step-by-step guides

Automated Updates to Software

- To implement the "*Automated Updates…*" control you might
  - Turn on auto updates for packages that support them
    - E.g. macOS, Windows & mobile OSs (ACSC Step-by-step guides)
    - Is your OS still supported? (https://endoflife.date/ Feb 2023)
      - iPhone 6, 5C, macOS 11 Big Sur, Android 10 Queen cake, ≈Win10
  - Use a package manager that automates updates, e.g.
    - Chocolatey, e.g. `DC Admin PS> choco upgrade -y all`
    - Winget, e.g. `PS> winget upgrade -all`

Later we will create a software inventory

ACSC Controls for Individuals and Families Advanced

	Examples	
Password manager [MFA]	• Protect with strongest passphrase • Ideally has MFA	
Ransomware protection	Controlled Folder Access in MS Win	Step-by-step guides
Secure IoTs	Isolate devices on guest WiFi	
Reduce digital footprint	• Delete unused accounts • Check social media privacy • Secure old emails	Step-by-step guides
Uninstall unused/non-reputable apps	Uninstall unused browser extensions	
Cyber emergency plan	Identify critical data	
Malware protection	Real time protection in MS Win	
Securely dispose mobiles/devices	Erase data, disassociate, remove memory cards & factory reset	
WiFi Router	Use WPA3	
Encrypt drive	BitLocker	

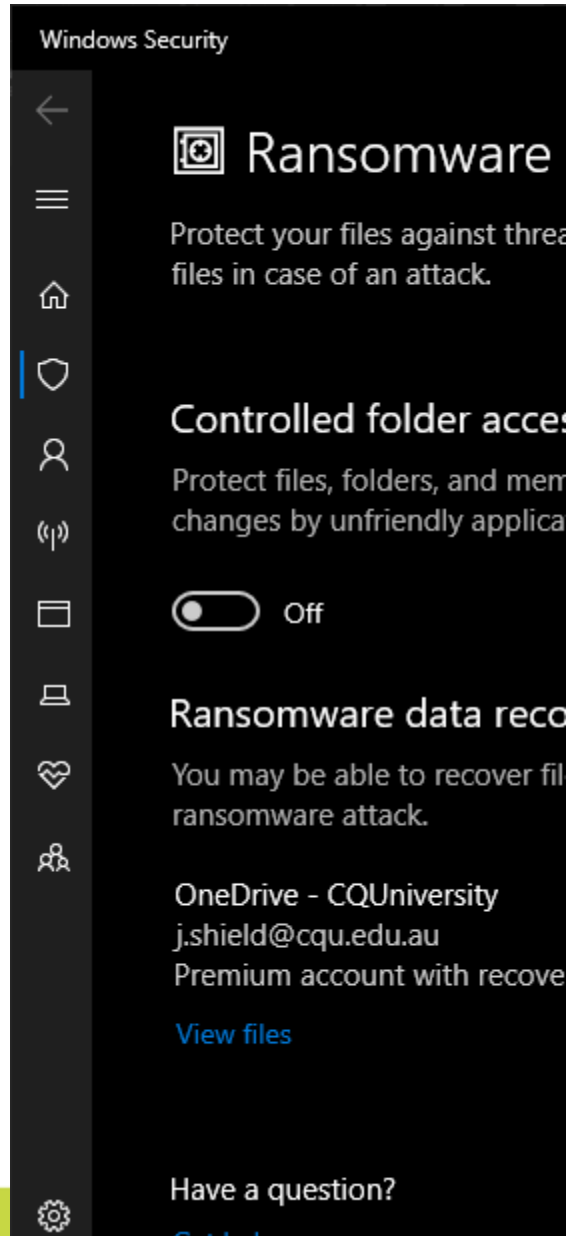
Extras

Controlled Folder Access is Ransomware Protection

- Blocks unauthorised apps from modifying protected files
- Enable Controlled Folder Access in Windows

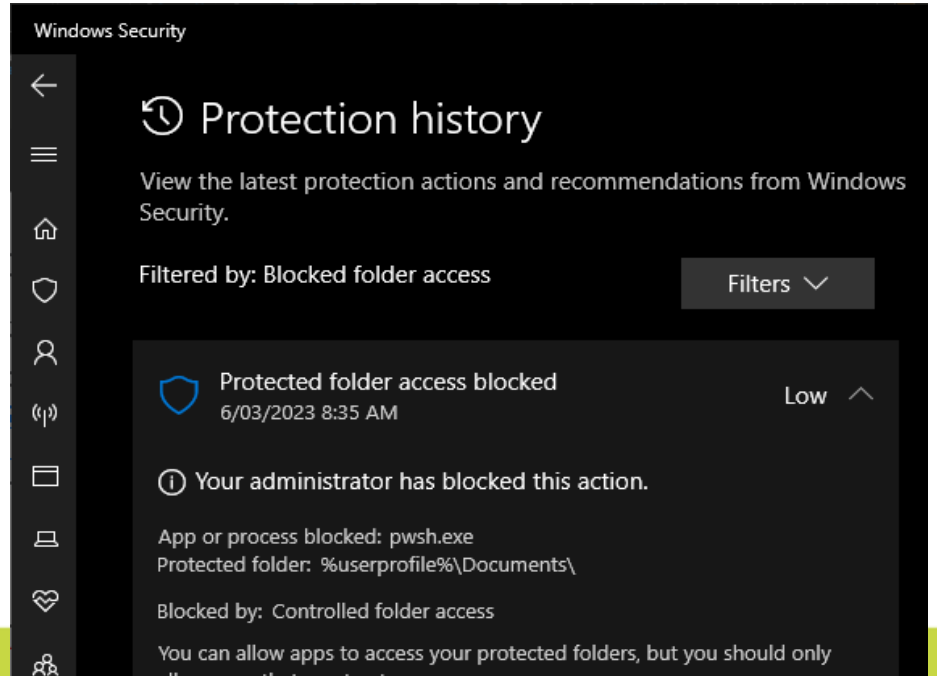
```
NotDC PS> Set-MpPreference -EnableControlledFolderAccess Enabled

NotDC PS> (Get-MpPreference).EnableControlledFolderAccess
1
```



Trigger Ransomware Protection

- Controlled Folder Access will block unauthorised apps from modifying protected files
  - Pwsh.exe not authorised
  - Folder $env:userprofile is protected
    - `PS> ls $env:userprofile`
- Simulate ransomware by writing to a protected file
  - `PS> echo "" > $env:userprofile\documents\test.txt`
    `Out-File: Could not find file 'C:\users\...\documents\test.txt'.`



Allow App through Controlled Folder Access

- Get path to app you wish to allow
  - ```
    PS> $pwsh=(Get-Command pwsh).source
    PS> echo $pwsh
    C:\Program Files\PowerShell\7\pwsh.exe
    ```
- Authorise app

```
PS> Add-MpPreference -ControlledFolderAccessAllowedApplications "$pwsh"
```

- Test

```
PS> echo "" > $env:userprofile\documents\test.txt
```
```

ACSC Essential Eight

- Top 8 controls that will do most to mitigate attacks on your enterprise, according to ACSC

Essential Eight	Examples
Implement app control	Limit executable apps, e.g. AppLocker
Patch apps	Daily vulnerability scan Remove unsupported apps
Restrict Office macros	E.g. only sandboxed, trusted locations or signed
Harden apps	No browser Java; no IE11 No Office or PDF child processes
Restrict admin privileges	Admin accounts blocked from internet
Patch OSs	Latest or previous version used for all devices
Implement MFA	Login attempts are logged
Regular backups	Only break-glass accounts can modify backups

cyber.gov.au

App Control mitigates Malware

- Only apps in an allowlist can execute
  - Windows tech, e.g. WDAC, Smart App Control(WDAC) & AppLocker (no longer preferred)
- DC has Win Defender App Control (WDAC) configured
  - Trigger WDAC by running non-approved app
    - `DC PS> keyfinder`
  - View alert in event log
    - `DC PS> eventvwr /c:"Microsoft-Windows-CodeIntegrity/Operational"`

Information	3/3/2023 4:30:53 PM	CodeIn...	3076	(18)

Event 3076, CodeIntegrity

General | Details

Code Integrity determined that a process (\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powershell.exe) attempted to load \Device\HarddiskVolume2\ProgramData\chocolatey\bin\keyfinder.exe that did not meet the Enterprise signing level requirements or violated code integrity policy (Policy ID:{a244370e-44c9-4c06-b551-f6016e563076}). However, due to code integrity auditing policy, the image was allowed to load.

WDAC Configuration

1. Define policy, e.g. list of permitted app sources / signatures
   - Only apps signed by authorised signers are allowed to execute
     - C:\Windows\schemas\CodeIntegrity\ExamplePolicies\DefaultWindows_Audit.xml
       - Audits, e.g. Windows Store apps
2. Compile policy XML, e.g.
   - Create a .cip file using `ConvertFrom-CIPolicy`
3. Deploy compiled policy to machine, e.g.
   - Using citool

https://learn.microsoft.com/en-us/windows/security/threat-protection/windows-defender-application-control/

Essential Eight Maturity Model

- Each of the essential 8 can be implemented with different maturity levels, e.g. for Application Control
  - Level 1 ≈ User PCs
    - App control of folders used by users, the OS, your web-browser & your email client
  - Level 2 ≈ Organisation
    - App control also implemented on servers
    - Organisation approves apps
    - Allowed & blocked execution events logged
  - Level 3, e.g.
    - Microsoft block rules implemented
    - Rules are validated at least annually
    - Execution events centrally logged, protected & monitored

https://www.cyber.gov.au/acsc/view-all-content/publications/essential-eight-maturity-model

CIS Controls

- Center for Internet Security critical security controls are broader than Essential 8
  - CIS Controls also prioritised to mitigate attacks

Essential Eight	Some CIS Controls v8
Implement app control	2 Inventory & control software
Patch apps	7 Continuous vulnerability management
Restrict Office macros	4 Secure Configuration
Harden apps	4 Secure Configuration
Restrict admin privileges	5 Account Management
Patch OSs	7 Continuous vulnerability management
Implement MFA	6 Access Control Management
Regular backups	11 Data Recovery

https://www.cisecurity.org/controls

More CIS Controls

- CIS Controls ≈ Essential 8 + ….

CIS Controls v8	Example
1 Inventory & control hardware	What devices are on the network?
3 Data Protection	Encryption
8 Audit Log Management	Collect events centrally, e.g. DNS lookups
9 Email & Browser Protections	Sender Policy Framework (SPF) ≈ approved senders
10 Malware Defenses	Defender App Guard ≈ MS Edge in container
12 Network Infrastructure Management	Segment network
…	…

CIS 1 Inventory and Control of Enterprise Assets

- 1.1 Establish & Maintain Detailed Enterprise Asset Inventory
  - Identify devices on network
    - Network discovery, e.g. Spiceworks (Chapman 2016, Chp 2), OpenAudit, ManageEngine, PRTG Network Monitor, WinAudit
      - Sniff traffic, e.g. Wireshark
      - DHCP logging
      - Vulnerability scanners, e.g. Nessus (≈Wk4 Vulnerabilities)
    - Active Directory ≈ Windows database of assets

```
DC PS> (Get-ADComputer -filter * ).DNSHostName
dc.windomain.local
win11.windomain.local
```

    - Asset management, e.g. ≈ MS Endpoint Manager

github.com_robvandenbrink_Critical-Controls-v7

CIS 1 Inventory and Control of Enterprise Assets

- 1.1 Establish & Maintain Detailed Enterprise Asset Inventory
  - Detailed inventory
    - Common Information Model (CIM)
      - Industry standard for device characteristics
      - Used by, e.g. MS & IBM

```
DC PS> Get-CimInstance CIM_ComputerSystem
Name Domain                TotalPhysicalMemory  Model        Manufacturer
DC   windomain.local       3220754432           VirtualBox   innotek GmbH
```

    - Wazuh Security Info & Event Management (SIEM) (≈Wk4)

Avoid WMI as it is deprecated

github.com_robvandenbrink_Critical-Controls-v7

CIS 1 Inventory and Control of Enterprise Assets

- 1.1 Establish & Maintain Detailed Enterprise Asset Inventory ✓
- 1.2 Address Unauthorised Assets
  - Identify unauthorised assets
    - IP address based
      - DHCP logging, e.g. Security Onion, Zeek (Bro)
      - Sniff traffic to identify IP, e.g. Wireshark

CIS 2 Inventory and Control of Software Asset

- 2.1 Establish & Maintain a Software Inventory
  - What software have you purchased?
  - Where are installation keys kept?
- 2.2 Ensure Authorised Software is Currently Supported
  - https://endoflife.date/
- 2.3 Address Unauthorised Software
- 2.4 Use Automated Software Inventory Tools
  - List (some) installed software
    - ```
      PS> Get-CimInstance CIM_Product
      Name                    Vendor
      Zoom                    Zoom
      Microsoft Search…       Microsoft…
      ```
- 2.5 Allowlist Authorised Software, e.g. WDAC
      ```

CIS Control Implementation Groups

- Analogous to Essential 8 Maturity Levels
  - IG1 ≈ Small business
  - IG2 ≈ IT Infrastructure & regulatory compliance
  - IG3 ≈ Security experts

	IG1	IG2	IG3
2.1 Establish & Maintain a Software Inventory	✓	✓	✓
2.2 Ensure Authorised Software is Currently Supported	✓	✓	✓
2.3 Address Unauthorised Software	✓	✓	✓
2.4 Use Automated Software Inventory Tools		✓	✓
2.5 Allowlist Authorised Software		✓	✓
2.6 Allowlist Authorised Libraries		✓	✓
2.7 Allowlist Authorised Scripts			✓

CIS Control 9 Email and Browser Protections

- 9.1 Use only supported browsers & clients
- 9.2 Use DNS filtering services
- 9.3 Maintain & enforce network based URL filters
- …
- 9.5 Implement DMARC

DNS Review

- Convert Domain Name Server (DNS) name to IP

```
PS> Resolve-DnsName www.duckduckgo.com
Name                Type   TTL   Section    NameHost
www.duckduckgo.com  CNAME  84024 Answer     duckduckgo.com
Name        : duckduckgo.com
QueryType   : A
TTL         : 23
Section     : Answer
IP4Address  : 20.43.111.112
```

- Query types include (`PS> help resolve-dnsname`)
  - A: IPv4
  - AAAA: IPv6
  - CNAME: Alias of Canonical name
    - Performs another DNS lookup

Deprecated C:> nslookup $ nslookup

$ host $ dig

DNS Review

- Domain name doesnotexist.dne does not exist

```
PS> Resolve-DnsName doesnotexist.dne
Resolve-DnsName : doesnotexist.dne : DNS name does not
exist
```

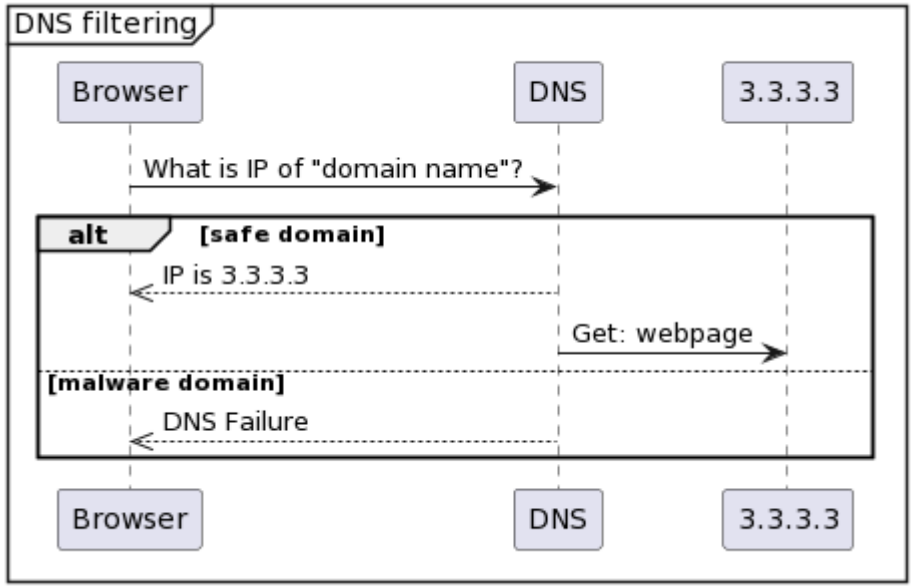
CIS Safeguard 9.2 DNS Filtering

- Browsers first resolve the domain name in a URL to an IP
- Some DNS servers have list of unsafe domain names to help prevent malware downloads, e.g.
  - Untrusted domain: 0-0-0.wakaf.net
  - Cloudflare DNS returns server failure

```
PS> resolve-dnsname 0-0-0.wakaf.net -server 1.1.1.3
resolve-dnsname : 0-0-0.wakaf.net : DNS server failure
```

  - Google DNS

```
PS> resolve-dnsname 0-0-0.wakaf.net
resolve-dnsname : 0-0-0.wakaf.net : DNS name does not exist
```



CIS Safeguard 9.5 DMARC

- CIS 9.5 recommends implementing SPF & DKIM
- Sender Policy Framework ≈ list of trusted email senders
- SPF – whitelist of servers permitted to send email on behalf of a domain

```
PS> resolve-dnsname _spf.google.com TXT -server 8.8.8.8
Name                Type Strings
_spf.google.com TXT  {v=spf1 include:_netblocks.google.com
                             include:_netblocks2.google.com
                             include:_netblocks3…
```

DomainKeys Identified Mail

- DKIM ≈ sign email to check for spoofing of mail senders
  - Sending-server forwards emails signed with its private key
  - Receiving-server verifies emails using Sending-server's public key
- Check headers of 1 of your emails, e.g. Email > Properties

  `a=rsa-sha256; s=arcselector9901; d=microsoft.com`
  - Obtain public key of server

```
PS> Resolve-DnsName arcselector9901._domainkey.microsoft.com TXT  -server 8.8.8.8
Name                                   Strings
arcselector9901._domainkey.microsoft.com {v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQ…
```

Domain Message Authentication Reporting and Conformance

- DMARC ≈ Hey, I've been receiving lots of emails from you
  - DMARC ≈ SPF + DKIM + reporting
  - Reporting
    - Receiver reports back to sender about an email
    - Sender makes suggestions to receiver, e.g. to quarantine or reject
- Where are reports sent?

```
PS> resolve-dnsname _dmarc.google.com TXT -server 8.8.8.8
Name                 Strings
_dmarc.google.com    {v=DMARC1; p=reject;
                      rua=mailto:mailauth-reports@google.com}
```

NIST CSF

- National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF)
  - Comprehensive, technology agnostic guidelines
  - NIST CSF ≈ Rearranged CIS Controls + more process, e.g.
    - Business Environment & Supply Chain Management

Essential Eight	Some CIS Controls v8	Some NIST CSF Controls
	1 Inventory & control hardware	ID.Physical asset management AM.1
Implement app control	2 Inventory & control software	ID.Software asset management AM.2
	3 Data Protection	PR.Data security DS
Restrict macros, harden apps	4 Secure Configuration	PR.Baseline configurations IP-1
Restrict admin privileges	5 Account Management	PR.Identity Management in AC
Implement MFA	6 Access Control Management	PR.Access control in AC
Patch apps & OSs	7 Continuous vulnerability management	≈DE.Security Continuous Monitoring CM
	8 Audit Log Management	PR.Audit logs PT.1
	9 Email & Browser Protections	
	10 Malware Defenses	DE.Malicious code detected CM
Regular backups	11 Data Recovery	PR.Backups IP-4

NIST CSF

- Organised around defenders' processes

Identify	Understand risks to assets, e.g. lose photos due to harddrive failure
Protect	Harden your setup, e.g. install firewall
Detect	Develop ways to identify an incident, e.g. IDS
Respond	Develop ways to react to an incident, e.g. turn off router & inform family
Recover	Develop ways to restore capabilities, e.g. restore from backup

NIST SP 800-53

- Controls ≈ design pattern for security requirements
  - Technology aware, e.g.
    - Off-load audit records due to limited audit storage
  - ≈20 Control Families including
    - AC: Access Control
    - IR: Incident response
    - MP: Media Protection
    - PE: Physical & Environment Protection
    - SC: protecting communications

(NIST 2020)

PowerShell Networking

- What are the known devices on your network segments?
  - PS> **Get-NetNeighbor**

```
PS C:\Users\shieldj> Get-NetNeighbor

ifIndex IPAddress                          LinkLayerAddress      State       PolicyStore
------- ---------                          -----------------     -----       -----------
133     224.0.0.251                        01-03-4E-02-01-FB     Permanent   ActiveStore
```

- What cache is being displayed?
  - Arp cache
- What does Permanent State mean?
  - Statically assigned IP address – not assigned dynamically using ARP

Deprecated arp

$ ip neighbour

PowerShell Networking

- Get gateway & DNS server
  - PS> **Get-NetIPConfiguration**

```
PS> Get-NetIPConfiguration
InterfaceAlias         : Ethernet
InterfaceIndex         : 6
InterfaceDescription   : Intel(R) Eth I12-LM
NetProfile.Name        : ad.cqu.edu.au
IPv4Address            : 138.77.110.176
IPv4DefaultGateway     : 138.77.110.1
DNSServer              : 138.77.6.15
```

Summary

	PS> Powershell	Deprecated
NIC MAC & link status	`Get-NetAdapter`	
NIC ARP (MAC) cache	`Get-NetNeighbor`	PS> arp -a
NIC IP	`Get-NetIPAddress`	PS> ipconfig
NIC Gateway/DNS Server	`Get-NetIPConfiguration`	
Test connection	`Test-NetConnection`	PS> ping
Resolve DNS to IP	`Resolve-DnsName`	PS> nslookup

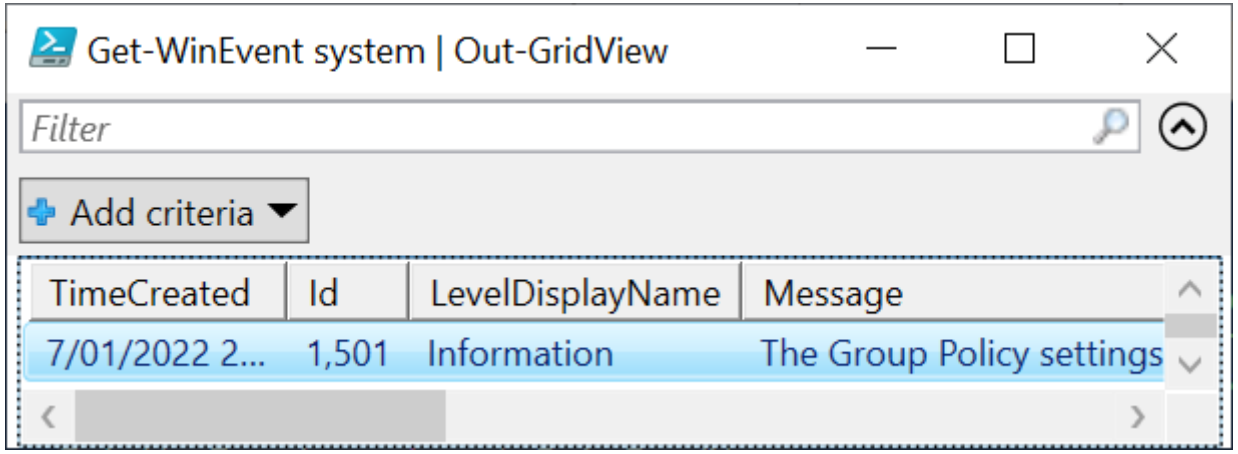
Events

- OS keeps logs of events such as loss of power
  - Motivation: search for abnormal events
- Start Windows' Event Viewer using Win + R > eventvwr or

```
DC PS> eventvwr
```

- Review events from system log
  - Messages categorised by Id
  - Newest message first

```
DC PS> Get-WinEvent system | Out-GridView
```



Events

- List apps blocked by WDAC
  - Event 3076 in log Microsoft-**Windows-CodeIntegrity/Operational**
  - DC **PS**> **(Get-WinEvent** "*CodeIntegrity*" | `
    Where-Object id **-eq** 3076 | `
    Select-Object -first 1).message
    Code **Int**egrity determined that a process (\Device\HarddiskVolume2\ProgramData\chocolatey\bin\keyfinder.exe)

Events

- List file modifications blocked by Controlled Folder Access
  - Event 1123 in "Microsoft-Windows-Windows Defender/Operational"
    -
```
NotDC PS> (Get-WinEvent "*Windows Defender*" | `
          Where-Object id -eq 1123 | `
          Select-Object -first 1).message
C:\...\pwsh.exe has been blocked from modifying
%userprofile%\Documents\ by Controlled Folder Access.
        Path: %userprofile%\Documents\
        Process Name: C:\Program Files\PowerShell\7\pwsh.exe
```

Won't work on DC

Events

- List other logs, e.g. DNS
  - Suffix 2>$null hides errors by redirecting them to null
  -
```
DC PS> Get-WinEvent -ListLog * 2>$null | more
LogMode   MaximumSizeInBytes RecordCount LogName
Circular            15728640        2811 Windows…

PS> (Get-WinEvent -ListLog "*DNS*" | Get-WinEvent).Message
The DNS server has started.
```

Common Information Model

- CIM has many tables

```
DC PS> Get-CimClass * | Out-GridView
CimClassName
------------
CIM_Battery
Win32_Battery
Win32_PortableBattery
CIM_Processor
CIM_LogicalDisk
CIM_DiskPartition
CIM_Memory …
```

WMI deprecated

Common Information Model

- Extract details from your PC
  - BIOS
    - ```
      DC PS> Get-CimInstance CIM_BIOSElement
      SMBIOSBIOSVersion : VirtualBox
      Manufacturer      : innotek GmbH
      ```
  - Processor
    - ```
      DC PS> Get-CimInstance CIM_Processor
      DeviceID Name                                    MaxClockSpeed
      CPU0     Intel(R) Core(TM) i5-8350U CPU @ 1.70GHz …
      ```
  - RAM
    - ```
      NotDC PS> (Get-CimInstance CIM_PhysicalMemory | `
                 measure -property capacity -sum).sum / 1gb
      16   (Reports 0 for VirtualBox)
      ```
      ```

Review Discover Commands

- Discover commands related to processes

```
PS> Get-Command -noun process

CommandType     Name                                   ...
-----------     ----
Cmdlet          Get-Process
Cmdlet          Stop-Process
...
```

- Cmdlets (commands), e.g. **Get-Date**, have a verb-noun structure to make them easier to learn

(Shepard et al. 2017, Mod 1 Chp 2)

Processes

- Motivation: search for abnormal processes
- Create a process to kill

```
PS> calc
PS> Get-Process Calculator
   Id ProcessName   CPU
22220 Calculator  1.625
```

- Kill all calculator processes

```
PS> man stop-process
 SYNTAX
     Stop-Process [-Id] <int[]> …
PS> Stop-Process (Get-Process Calculator).Id
```

```
xcalc &
```

```
Get-Process xcalc
```

alias `Get-Help`

```
Stop-Process (Get-Process xcalc).Id
```

Bitcoin Processes

- Identify top 2 processes which use your CPU
  - You think you have a bitcoin miner
  - Omit units (s) when specifying property names such as CPU

```
PS> Get-Process | `
    Sort-Object -property CPU -descending | `
    Select-Object -first 2

   Id ProcessName        CPU
   -- -----------        ---
 7144 OneDrive            449
10772 EMMeter     269.984375
```

Services

- Motivation: monitor firewall service for abnormal activity
  - PS> **Get-Command** -Noun Service
    Cmdlet **Get-Service** ...
    PS> **man Get-Service**
    Get-Service -DisplayName <**string**[]> ...
    PS> **Get-Service** | **Out-Gridview**
    Status Name DisplayName
    Stopped ALG Application Layer Gateway…
  - Is Windows Defender Firewall service running?
    - PS> **(Get-Service** | `
      Where-Object DisplayName -like "*Defender Firewall*").Status
      Running

(Shepard et al. 2017, Mod 3 Chp 12)

Web Services

- Motivation: check remote servers
- Obtain list of songs from iTunes

Alias **Invoke-WebRequest**

```
PS> $Json=(iwr -uri "https://itunes.apple.com/search?term=astley").Content
PS> $Json
```

- JSON is a format for serialising objects, i.e. for storing objects in text files

```
– StatusCode        : 200
– StatusDescription : OK
– Content           : {"resultCount":50,
–                      "results": ["kind":"song", "artistName":"New Kids …
```

– Convert $Json to object

```
PS> ($Json | ConvertFrom-Json).Results | Out-GridView
 artistName                : New Kids on the Block
 trackName                 : Bring Back the Time
 ...
```

(Shepard et al. 2017, Mod 2 Chp 4)

Environment Variables

- env: “drive” stores environment variables
  - PS> **ls** env:

```
Name     Value
TEMP     C:\Users\shieldj\AppData\Local\Temp
USERNAME shieldj
windir   C:\WINDOWS
```

  - PS> **echo** ${env:TEMP}

```
C:\Users\shieldj\AppData\Local\Temp
```

```
C:> set
$ env
```

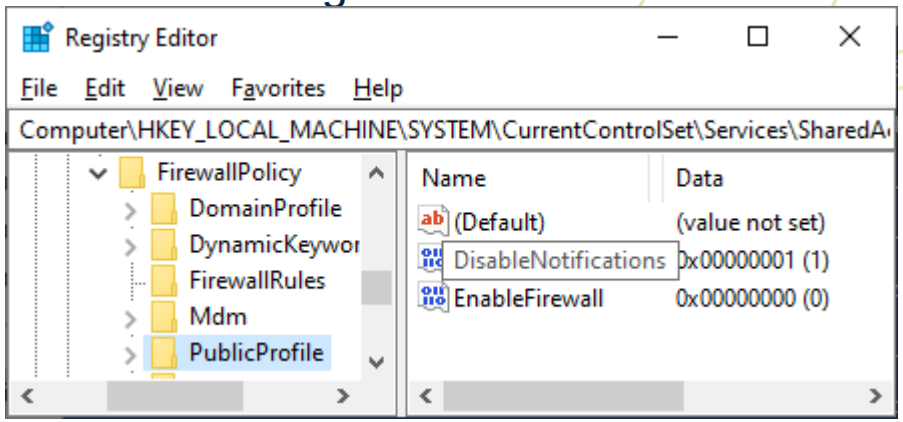
```
C:> echo %TEMP%
$ echo ${TEMP}
```

- New temporary files are stored in $env:TEMP
  - PS> **New-TemporaryFile** | **Select** Name,DirectoryName

```
Name         DirectoryName
tmpD771.tmp C:\Users\shieldj\AppData\Local\Temp
```

Registry

Dangerous to edit



- Configuration settings store
  - E.g. HKLM "drive"
  - "Folders" contain properties

```
DC PS> ls "HKLM:\SYSTEM\CurrentControlSet\Services\SharedAccess\Parameters\FirewallPolicy"
Name                                    Property …
PublicProfile                           DisableNotifications : 1
                                        EnableFirewall       : 0 …
```

  - List properties

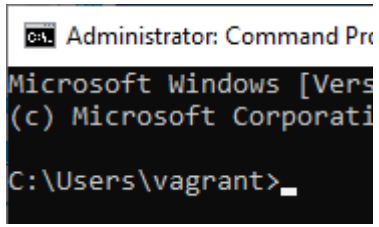
```
PS> Get-ItemProperty "HKLM:\SYSTEM\CurrentControlSet\Services\SharedAccess\Parameters\FirewallPolicy\PublicProfile"
EnableFirewall          : 0 …
```

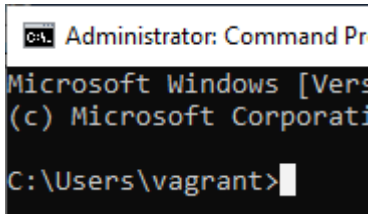
  - Meanings
    - 0 = Firewall disabled; 1 = Enabled

Registry

- Change your console cursor size using the registry

```
DC PS> ls HKCU:
    Hive: HKEY_CURRENT_USER
Name          Property                          
Console       ColorTable00                  : 789516
              CursorSize                    : 25
DC PS> set-itemproperty HKCU:\Console CursorSize 100
```





References

Happy Studying!

Chapman, C 2016, *Network Performance and Security: Testing and Analyzing Using Open Source and Low-Cost Tools*. Rockland, MA: Elsevier Science & Technology Books. https://cqu.primo.exlibrisgroup.com/permalink/61CQU_INST/669qn3/cdi_skillsoft_books24x7_bks000112765

Kohnke, A, Shoemaker, D & Sigler, E 2016. *The Complete Guide to Cybersecurity Risks and Controls*, Auerbach Publications.

NIST 2020 *NIST Special Publication 800-53 Rev 5: Security and Privacy Controls for Information Systems and Organisations*, https://doi.org/10.6028/NIST.SP.800-53r5

Shepard, M, Venkatesan, C, Talaat, S & Blawat, B 2017. *PowerShell,* 1st edn, Packt, viewed 17 January 2022, https://cqu-primo.hosted.exlibrisgroup.com/permalink/f/1rb43gr/TN_cdi_igpublishing_primary_PACKT0000750